2019-Summer Paper 3 – Subspeciality of Urology

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Q1. Discuss differential diagnosis of small renal tumours. Describe techniques to achieve hemostasis during laparoscopic partial nephrectomy.

Type: LAQ (25 marks)

Definition and biology

- **SRM (small renal mass):** an enhancing solid or complex cystic renal mass **≤4 cm (cT1a)**, usually found incidentally.
- **Benign share falls with size:** 20–30% of masses ≤4 cm and about 40% of masses ≤2 cm are benign. At ≥7 cm few are benign and about half are high grade.
- Male sex triples the risk of malignancy. Under 2% of small cancers metastasise.

Differential diagnosis

Group	Entities	Pointers
RCC (renal cell carcinoma)	**Clear cell** (commonest), papillary (10–15%), chromophobe (3–5%), cystic RCC, rare molecular subtypes	Clear cell enhances strongly; papillary enhances weakly
Other malignant	Urothelial carcinoma, lymphoma, metastasis, sarcoma, epithelioid AML (angiomyolipoma)	Infiltrative growth
Benign solid	Oncocytoma, fat-poor AML, papillary adenoma, metanephric adenoma	See below
Cystic	Haemorrhagic or complex cyst, cystic nephroma	Bosniak category
Inflammatory, vascular	Abscess, focal pyelonephritis, tuberculosis; aneurysm, AVM (arteriovenous malformation), infarct, pseudotumour	Clinical picture

- **Oncocytoma:** commonest benign enhancing SRM (up to half of benign SRMs). Central stellate scar and segmental enhancement inversion suggest it but are not specific. A core cannot separate it from chromophobe RCC, which has oncocytic areas. Sestamibi SPECT (single-photon emission computed tomography) is under evaluation.
- **Fat-poor AML:** no visible fat (fat below −20 HU [Hounsfield units] diagnoses classic AML); homogeneous, T2-dark, mimics RCC; biopsy HMB-45 and Melan-A positive.
- **Also know:** metanephric adenoma (hypovascular, polycythaemia, WT1 positive); papillary adenoma (≤15 mm).

Narrowing the differential

- **Multiphase CT (computed tomography):** enhancement over 15–20 HU means solid tissue. MRI (magnetic resonance imaging) helps below 2 cm.
- **RMB (renal mass biopsy)** if it will change management (AUA [American Urological Association]); seeding almost unreported; non-diagnostic about 14%.
- Then surveillance, ablation or **PN (partial nephrectomy)**, preferred for cT1a.

Haemostasis in LPN (laparoscopic partial nephrectomy)

Principle: a clear field for excision, then secure closure of the bed. Plan with R.E.N.A.L. score and 3D (three-dimensional) CT; intraoperative ultrasound.

1. Vascular control

Technique	Method	Comment
Main artery ± vein clamp	Bulldog clamps, or laparoscopic Satinsky en bloc	Artery-only equals artery–vein. Aim warm ischaemia ≤25 min (trifecta)
Selective or super-selective	Clamp the tumour's segmental branch; 3D CT, NIRF (near-infrared fluorescence)	Avoids global ischaemia; no functional gain in an RCT (randomised controlled trial)
Off-clamp ("zero ischaemia")	No clamping; small exophytic tumours	More blood loss; same long-term eGFR (estimated glomerular filtration rate)
Early unclamping	Release after the inner suture layer	Shorter ischaemia; possibly fewer pseudoaneurysms
Cold ischaemia	Ice slush, cold saline via ureteric catheter, cold arterial perfusion	Solitary kidney, CKD (chronic kidney disease), long repair; benefit unproven

- Function depends mainly on parenchyma preserved. Mannitol: no benefit in an RCT. **Also know:** parenchymal clamp for peripheral tumours.

2. Excision: score the capsule, cut with cold scissors, clip visible vessels. Enucleation along the pseudocapsule bleeds less.

3. Renorrhaphy (most reliable step)

- **Inner layer:** running 3-0 or 2-0 absorbable suture closes cut vessels and collecting system (opened in about 60%).
- **Outer layer, sliding-clip technique:** mattress sutures anchored by a locking clip; clips slid down to the capsule give tension without knots. Pledgets prevent capsular tears; oxidised cellulose bolster fills the defect.
- **Barbed suture:** holds tension without knots, so ischaemia is shorter.
- Single layer gave better function and fewer pseudoaneurysms than two layers.

4. Energy: argon beam or bipolar coagulation of the bed. Argon can destroy collecting-system sutures and adds gas, so vent the ports.

5. Haemostatic agents (adjuncts, no proven benefit in RCTs): gelatin–thrombin matrix; fibrin sealant; oxidised cellulose; PEG (polyethylene glycol) hydrogel; cyanoacrylate. Sutureless closure only for selected small tumours.

6. Final check: unclamp, inspect, **lower pneumoperitoneum** to unmask venous bleeding; drain.

Postoperative bleeding

- Early (hypotension, falling haemoglobin): resuscitate, CT, angioembolisation.
- **Delayed haematuria (about day 12):** pseudoaneurysm or arteriovenous fistula; selective angioembolisation preserves function. Endophytic and posterior tumours carry higher risk.

Diagram to draw: sliding-clip renorrhaphy over a bolster, with bulldog clamp on the artery.

Key points to remember:

- SRM ≤4 cm: 20–30% benign; ≤2 cm about 40%.
- Oncocytoma is the commonest benign mimic; biopsy cannot exclude chromophobe RCC.
- Trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Delayed haematuria after PN means pseudoaneurysm: angioembolise.

Q2. Describe evaluation of SUI in females. Discuss surgical therapy of SUI in Males.

Type: LAQ (25 marks)

Part A: Evaluation of SUI (stress urinary incontinence) in women

- **Definition (ICS [International Continence Society]):** involuntary leakage on effort, exertion, sneezing or coughing.
- **Mechanisms:** urethral **hypermobility** (loss of suburethral support, DeLancey hammock) and **ISD (intrinsic sphincter deficiency)**. Most women have some ISD.

History

- Leakage with cough or activity means SUI; after urgency, UUI (urgency urinary incontinence); both, MUI (mixed). A positive SUI history predicts urodynamic SUI in about three-quarters.
- Severity (pads/day), bother; voiding symptoms, prolapse, obstetric history, prior surgery or radiation, neurological disease, drugs.
- **Questionnaires:** ICIQ-UI SF (International Consultation on Incontinence Questionnaire–UI Short Form); UDI-6 (Urogenital Distress Inventory: bother); IIQ-7 (Incontinence Impact Questionnaire: impact). **Bladder diary** (3 days).

Examination (comfortably full bladder)

- Neurological (S2–S4 sensation, anal tone, bulbocavernosus reflex); atrophy; prolapse by **POP-Q (Pelvic Organ Prolapse Quantification)**.
- **Cough stress test:** cough supine, then standing if negative. Leakage synchronous with cough confirms SUI. Reduce prolapse to unmask **occult SUI**.
- **Q-tip test:** swab to bladder neck, Valsalva; rotation **>30°** = hypermobility. Visual mobility examination is now routine.

Tests

Test	Method and use
Urinalysis	Required; exclude infection, haematuria
PVR (post-void residual)	Required before surgery; ultrasound; repeat if high (no agreed cut-off)
1-hour pad test (ICS)	500 mL fluid, standard exercises; >1.4 g positive; 1–10 g mild, 11–50 g moderate, >50 g severe
24-hour pad test	Normal activities; more reproducible
Cystoscopy	Haematuria, irritative symptoms, previous mesh

Urodynamics (UDS): not routine. VALUE trial (2012): office evaluation was non-inferior to UDS before surgery for uncomplicated SUI.

- **Indications (AUA [American Urological Association]/SUFU [Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction] 2023):** unclear diagnosis or SUI not shown; urgency-predominant MUI; neurogenic bladder; high PVR or voiding dysfunction; prolapse stage ≥3 without demonstrable SUI; failed previous surgery.
- **ALPP/VLPP (abdominal/Valsalva leak point pressure; McGuire 1993):** <60 cmH2O = ISD; 60–90 equivocal; ≥90 = little ISD. A low MUCP (maximum urethral closure pressure, <20 cmH2O) also suggests ISD.

Part B: Surgical therapy of male SUI

- **Aetiology:** commonest after **radical prostatectomy (RP)**, which removes the internal sphincter, leaving continence to the rhabdosphincter. Also TURP (transurethral resection of the prostate), radiation, pelvic fracture.
- **Work-up:** 24-h pad weight (mild <150 g, moderate <400 g, severe above); standing cough test; **cystoscopy** (anastomotic stricture, sphincter coaptation); UDS if mixed or poor contractility.
- **Timing:** PFMT (pelvic floor muscle training) first; continence improves for up to 2 years, but do not delay surgery beyond 6 months for severe leak.

Devices

Device	Mechanism	Use and results
Transobturator sling (AdVance XP)	Repositions the bulb 3–4 cm proximally; needs residual sphincter	Mild–moderate, no radiation; cure about 60%
Quadratic sling (Virtue)	Compression plus elevation	Mild; limited data
Adjustable sling (ATOMS, adjustable transobturator male system)	Cushion adjusted via scrotal port	Mild–moderate; about two-thirds dry
ProACT balloons	Bladder-neck compression	More erosion
Bulking agents	Submucosal bulk	Small leaks, unfit men; rarely durable
AUS (artificial urinary sphincter)	Circumferential compression	**Gold standard**; all severities, after radiation

AMS 800 AUS

- **Components:** urethral **cuff** (bulbar, usually 4.0–4.5 cm), **PRB (pressure-regulating balloon) 61–70 cmH2O** (51–60 high risk; 71–80 bladder neck) in a preperitoneal pocket, scrotal **pump**. Filled with 23 mL.
- **Technique:** perineal incision; cuff on proximal bulbar urethra; **deactivate for 4–6 weeks**.
- **Needs** dexterity and cognition. Social continence (0–1 pad) 55–77%; satisfaction about 90%; lifespan 7–10 years.
- **Complications:** infection 1–3% (explant; reimplant after 3–6 months); **erosion** 2–15% (explant, catheter about 3 weeks); urethral **atrophy** (late recurrent leak); mechanical failure; retention.
- **Erosion risks:** radiation, prior erosion, urethroplasty; use **transcorporal cuff** (tunica adds bulk) or lower PRB. Atrophy: smaller or tandem cuff.

Algorithm (AUA/SUFU 2019)

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Severe SUI or any radiotherapy -> AUS
Moderate, no radiotherapy      -> AUS or male sling
Mild, no radiotherapy          -> sling (or PFMT, AUS)
Sling failure                  -> AUS
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- **MASTER RCT (randomised controlled trial):** sling was **non-inferior** to AUS for continence at 12 months; AUS remains preferred for heavy leak (>250 g/day) or radiation.

Diagram to draw: AMS 800 (cuff, pump, PRB) and transobturator sling path.

Key points to remember:

- Q-tip >30° = hypermobility; VLPP <60 = ISD.
- VALUE trial: UDS not needed in uncomplicated SUI.
- AUS PRB 61–70 cmH2O; deactivate 4–6 weeks.
- Radiation or severe SUI: AUS first; MASTER: sling non-inferior.

Q3a. Prepucioplasty

Type: SAQ (10 marks)

Read as: preputioplasty, a foreskin-preserving operation for phimosis.

Definition

- An operation that widens the tight preputial ring but keeps the foreskin. It is an alternative to circumcision.

Background

- **Physiological phimosis** at birth is due to glans–preputial adhesions. Smegma and erections separate them. About 90% retract by 3 years; fewer than 1% are phimotic at 17 years.
- **Secondary phimosis** follows scarring: forced retraction, balanoposthitis or BXO (balanitis xerotica obliterans).

Indications

- Symptomatic phimosis: recurrent balanoposthitis, UTI (urinary tract infection), paraphimosis, painful erection.
- Failed **topical steroid** (betamethasone 0.05% twice daily for about 4 weeks with gentle retraction). About 1 in 6 fail.
- Wish to keep the foreskin (cultural reasons, cosmesis).

Contraindications

- **BXO** with scarring: circumcision is preferred. Triradiate incision with triamcinolone has been reported in selected boys.
- Physiological non-retractility needs only reassurance. Treat active infection first.

Techniques

Technique	Method
Dorsal slit with transverse closure	Longitudinal incision through the ring, closed transversely (Heineke–Mikulicz principle)
Triple incision	Three longitudinal cuts over the ring, each closed transversely
Y–V plasty	V-flap of outer skin advanced into a Y-cut

- **Principle:** cutting the ring lengthwise and closing it crosswise widens its circumference.
- General anaesthesia or penile block; incise only the band; confirm retraction; absorbable sutures.

Postoperative care

- **Daily retraction from the first week**, continued for weeks, prevents re-stenosis. A topical steroid may help.

Outcomes and complications

- Most achieve a retractile foreskin. It is quicker and less painful than circumcision.
- Complications: re-phimosis (needs circumcision), bleeding, infection, poor cosmesis, paraphimosis.
- **Also know:** preputial reconstruction during hypospadias repair is also called preputioplasty; dehiscence is its risk.

Diagram to draw: dorsal incision through the ring, closed transversely.

Key points to remember:

- Steroid first; about 1 in 6 fail.
- Cut lengthwise, close crosswise; daily retraction after.
- BXO: circumcision preferred.

Q3b. Baceriuria and use of antibiotic policy in pregnancy

Type: SAQ (10 marks)

Read as: bacteriuria in pregnancy and the antibiotic policy for it.

Definition

- **ASB (asymptomatic bacteriuria):** $\geq 10^5$ CFU (colony-forming units)/mL of one organism without urinary symptoms.

Why pregnancy matters

- Prevalence is **not raised** (about 2–7%), but ASB **persists and ascends**.
- **Changes:** progesterone relaxes ureteric muscle and the uterus compresses the ureters at the pelvic brim (right more), causing stasis and hydroureteronephrosis. GFR (glomerular filtration rate) rises 30–50%, lowering drug levels.
- Untreated ASB led to **pyelonephritis** in 15–45% (historical), mostly in the third trimester, causing **preterm birth and low birth weight**.
- **Cochrane 2019:** treatment reduces pyelonephritis, preterm delivery and low birth weight.

Screening

- **Urine culture at 12–16 weeks** or first visit (USPSTF [US Preventive Services Task Force]); dipstick misses cases.

Antibiotic policy

Safe	Restricted or avoided
Penicillins (amoxicillin), cephalosporins: throughout	**Nitrofurantoin:** stop near term (from 35 weeks): neonatal haemolysis (G6PD [glucose-6-phosphate dehydrogenase] deficiency)
Fosfomycin: cystitis only	**TMP-SMX (trimethoprim-sulfamethoxazole):** avoid 1st trimester (folate, neural tube); caution near term (kernicterus)
Gentamicin: pyelonephritis	**Fluoroquinolones** (cartilage), **tetracyclines** (teeth, bone), chloramphenicol: avoid

- ACOG (American College of Obstetricians and Gynecologists) 2017: nitrofurantoin and sulfonamides first-line in 2nd–3rd trimester; 1st only if no alternative.
- **ASB or cystitis:** culture-directed, **4–7 days** (cures more than single dose); **repeat culture** after treatment.

Pyelonephritis

- **Admit**; IV (intravenous) ampicillin plus gentamicin, ceftriaxone or aztreonam, then oral cephalexin; 10–14 days. No improvement by 72 h: renal ultrasound.

Prophylaxis

- After one pyelonephritis or two cystitis episodes: nightly nitrofurantoin or cephalexin until delivery (Cochrane 2015: close surveillance equally effective).
- **Also know:** diversion, augmentation, SCI (spinal cord injury): treat ASB, consider prophylaxis.

Diagram to draw: right hydroureter at the pelvic brim.

Key points to remember:

- Screen by culture at 12–16 weeks; treat all ASB.
- β-lactams safe throughout; no fluoroquinolones or tetracyclines.
- Nitrofurantoin not near term; TMP not in 1st trimester.

Q3c. Accessories for RIRS

Type: SAQ (10 marks)

Introduction

- RIRS (retrograde intrarenal surgery) is flexible ureteroscopy in the kidney. Its single **3.6 Fr channel** is shared by accessories and irrigation.

Guidewires (0.018–0.038 inch, 145–260 cm)

Type	Construction	Use	Weakness
Hydrophilic	Nitinol core, slippery when wet	Bypassing impacted stones, tortuous ureter	Displaces easily
PTFE (polytetrafluoroethylene)-coated	Steel or nitinol	Safety and working wire	Damages scope channel on backloading
Hybrid	Floppy hydrophilic tip, stiff PTFE shaft	Both roles	—

- **Safety wire** keeps access and allows stenting after injury (AUA [American Urological Association], EAU [European Association of Urology]).
- **Also know:** open-ended or dual-lumen catheter (pyelogram, second wire); stiff wire for the sheath; balloon dilators.

UAS (ureteral access sheath)

- Hydrophilic sheath with tapered obturator; usually **12/14 Fr** (inner/outer); tip below the PUJ (pelviureteric junction).
- **Pros:** lower intrarenal pressure and temperature; repeated passes; less scope wear.
- **Cons:** ureteric injury in nearly half, severe in about **1 in 8** (less if pre-stented); no proven fall in infection.
- **Suction sheaths and FANS (flexible and navigable suction access sheath):** aspirate dust and keep pressure low.

Retrieval devices

- **Nitinol tipless baskets** (about 1.5–2.4 Fr): kink-resistant, open flush in calyces; relocate lower-pole stones. Small baskets allow more flow but are fragile. Graspers release easily.

Laser and fibres

- **Ho:YAG (holmium:yttrium-aluminium-garnet)** or **thulium fibre laser**; small-core fibres (about 200 µm) keep deflection and flow; **≤10 W in the ureter**.

Irrigation and antiretropulsion

- Gravity, pressure bag, hand or foot pumps (hand pumps: more fever); pressure sensors emerging.
- Stone Cone, NTrap, BackStop gel: mainly ureteric stones.

Stents

- **Double-J**; AUA: may omit after uncomplicated URS (ureteroscopy).

Diagram to draw: sheath over wire, tipless basket in lower calyx.

Key points to remember:

- Hydrophilic wire passes; PTFE wire stays.
- 12/14 Fr sheath lowers pressure; severe injury about 1 in 8.
- Nitinol tipless basket for calyces; ≤10 W in ureter.

Q3d. Immediate post TURBT intravesical Therapy for TCC

Type: SAQ (10 marks)

Introduction

- After TURBT (transurethral resection of bladder tumour), NMIBC (non-muscle-invasive bladder cancer) recurs in 40–60%. Many early recurrences come from **floating cells implanting** on the raw bed.
- A single chemotherapy instillation (SI) kills them before implantation. TCC (transitional cell carcinoma) is now called urothelial carcinoma.

Rationale and timing

- Implantation occurs within hours, so instil **as early as possible, ideally within 2–6 h** (theatre or recovery). Guidelines allow up to 24 h, but MMC given at 24 h lost its benefit.

Agents

Agent	Dose	Note
MMC (mitomycin C)	40 mg in 20–40 mL	Alkylating; skin rash
Gemcitabine	2 g in 100 mL, 1 h	Standard in USA; mild toxicity
Epirubicin	50 mg	Chemical cystitis

- **Also know:** pirarubicin works; thiotepa does not.

Technique

- Exclude perforation; instil via catheter; retain 1–2 h; drain.

Evidence

- **Sylvester 2016 individual-patient meta-analysis:** about 35% relative reduction in recurrence; **no benefit** with >1 recurrence/year or EORTC (European Organisation for Research and Treatment of Cancer) recurrence score ≥5.
- **SWOG (Southwest Oncology Group) S0337:** gemcitabine versus saline in suspected low-grade NMIBC: 4-year recurrence 35% vs 47%; no excess toxicity.
- No effect on progression.

Who benefits

- **Low risk** (primary solitary low-grade Ta ≤3 cm): SI alone; greatest benefit.
- Intermediate risk: SI then adjuvant course. High risk: BCG (bacille Calmette-Guérin) follows.

Contraindications

- **Suspected perforation**, extensive resection, bleeding needing irrigation.
- **Never BCG immediately**: intravasation of live bacteria causes sepsis and death.

Complications

- Irritative symptoms (commonest), chemical cystitis, skin desquamation (MMC), contracted bladder; serious harm mainly after perforation.
- **Also know:** continuous saline irrigation may also reduce recurrence.

Diagram to draw: floating cells implanting on resection bed → killed by instillation.

Key points to remember:

- Instil within 6 h (at most 24 h); never BCG.
- SWOG S0337: 35% vs 47% recurrence with gemcitabine 2 g.
- Withhold if perforation; best for low-risk solitary tumours.

Q3e. Endoscopic correction of VUR : indications and complications

Type: SAQ (10 marks)

Introduction

- A **bulking agent** injected under the ureteric orifice supports and lengthens the intramural tunnel, restoring the flap-valve in VUR (vesicoureteral reflux).
- First done with Teflon (1981). **Dx/HA (dextranomer/hyaluronic acid, Deflux)** is now standard.

Agents

Agent	Property	Problem
Teflon (PTFE [polytetrafluoroethylene] paste)	Permanent, particles <40 µm	Migration; never FDA (US Food and Drug Administration) approved
Dx/HA	Microspheres 80–250 µm; gel resorbed	Volume loss about 23%; late recurrence
Polyacrylate (Vantris)	Permanent, 300 µm	Late ureteric obstruction

Technique

Method	Needle entry
STING (subureteric Teflon injection)	2–3 mm below orifice, at 6 o'clock
HIT (hydrodistension implantation)	Inside the distended orifice; better than STING
Double HIT	Two intraureteric sites; about 90% resolution

- Aim for a **"volcano" mound** with a slit-like orifice.

Indications

- Persistent primary VUR (grades II–IV) with **breakthrough febrile UTI** (urinary tract infection) on prophylaxis, **new scars**, poor adherence or parental preference.
- Older girls; residual VUR after reimplantation; duplex or secondary VUR (lower success).
- **Relative contraindications:** grade V, **UVJ (ureterovesical junction) obstruction**, untreated BBD (bladder and bowel dysfunction), large diverticulum.

Success

- **Elder 2006 meta-analysis** (one injection): grade I–II about 79%, III 72%, IV 63%, V 51%; repeat injections raise success.
- **Swedish Reflux Trial:** 20% recurrence at 2 years after initial success.

Complications

- **Failure or recurrence** (commonest): repeat injection (1–2), then reimplantation.
- **New contralateral VUR** after unilateral injection.
- **Ureteric obstruction:** early or **late-onset**, so long-term ultrasound.
- Febrile UTI despite negative VCUG (voiding cystourethrogram).
- **Calcified mound** (over half) mimics a ureteric stone. Teflon migration.

Follow-up

- Day case; antibiotics 3 months; ultrasound at 2–6 weeks; ultrasound ± VCUG at 3 months.

Diagram to draw: STING versus HIT needle positions with volcano mound.

Key points to remember:

- Dx/HA 80–250 µm; HIT better than STING.
- Elder 2006: grade IV 63% after one injection.
- Watch for late obstruction and calcified mound.

Q3f. Management of DSD

Type: SAQ (10 marks)

Read as: disorders/differences of sex development. Detrusor–sphincter dyssynergia is covered in 2016-Summer Paper 1, Q3a.

Definition and classification (Chicago consensus 2006)

- Congenital conditions with atypical chromosomal, gonadal or anatomical sex.

Group	Examples
46,XX DSD	**CAH (congenital adrenal hyperplasia)**, 95% 21-hydroxylase; ovotesticular; testicular (SRY+ [sex-determining region Y]); maternal androgens
46,XY DSD	Gonadal dysgenesis; androgen synthesis defects (5α-reductase); androgen insensitivity (CAIS/PAIS, complete/partial)
Sex chromosome DSD	45,X/46,XY mixed gonadal dysgenesis; 45,X Turner; 47,XXY Klinefelter

Neonatal evaluation (first 24–48 h)

- **Emergency:** exclude **salt-wasting CAH** (low sodium, high potassium, shock; usually after day 5–7).
- **MDT (multidisciplinary team):** urologist, endocrinologist, geneticist, psychologist, ethicist; gender-neutral language.
- **Examination:** stretched phallic length, meatus, labioscrotal fusion (Prader stage), **gonads**. Impalpable gonads suggest CAH; a palpable gonad means testicular tissue.
- **Tests:**
  - **Karyotype** (FISH [fluorescence in situ hybridisation] for SRY in hours).
  - **17-OHP (17-hydroxyprogesterone)** after 36 h; electrolytes, glucose.
  - LH (luteinising hormone), FSH (follicle-stimulating hormone), testosterone, AMH (anti-Müllerian hormone); testosterone:DHT (dihydrotestosterone) >20:1 = 5α-reductase deficiency; hCG (human chorionic gonadotropin) stimulation.
  - **Pelvic ultrasound** for uterus; then genitography, endoscopy, laparoscopy ± biopsy, genetic panels. No gonadectomy while diagnosing.

Medical management

- **CAH:** hydrocortisone, fludrocortisone, salt; stress dosing. **Puberty:** oestrogen or testosterone induction if gonads fail.

Gender assignment

- Weighs diagnosis, androgen exposure, fertility, sexual function, family wishes. Usually female in CAH and CAIS; male trend in 5α-reductase deficiency.

Surgery: timing and consent

- **Feminising** surgery (clitoroplasty, vaginoplasty, labioplasty) is irreversible, so deferral until the child can consent is favoured unless obstructed; severe CAH is debated.
- **Masculinising** (proximal hypospadias repair, orchidopexy) is often done in infancy.

Gonadal tumour risk

Risk	Condition	Management
High	Dysgenesis with Y, intra-abdominal (15–35%); PAIS non-scrotal (50%)	Gonadectomy

Risk	Condition	Management
Intermediate	Turner with Y (12%)	Gonadectomy
Low	CAIS (2%); ovotesticular (3%)	CAIS: keep until after puberty; ovotesticular: remove discordant tissue

Diagram to draw: karyotype-based DSD algorithm.

Key points to remember:

- 17-OHP after 36 h; salt crisis after day 5–7.
- Palpable gonad = testicular tissue.
- Defer irreversible surgery; CAIS gonads kept till puberty.

Rapid Revision Sheet – 2019-Summer Paper 3

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Discuss differential diagnosis of small renal tumours. Describe techniques to achieve hemostasis during laparoscopic partial nephrectomy. (LAQ)

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- VALUE trial: UDS not needed in uncomplicated SUI.
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- Radiation or severe SUI: AUS first; MASTER: sling non-inferior.

Q3a. Prepucioplasty (SAQ)

- Steroid first; about 1 in 6 fail.
- Cut lengthwise, close crosswise; daily retraction after.
- BXO: circumcision preferred.

Q3b. Baceriuria and use of antibiotic policy in pregnancy (SAQ)

- Screen by culture at 12–16 weeks; treat all ASB.
- β-lactams safe throughout; no fluoroquinolones or tetracyclines.
- Nitrofurantoin not near term; TMP not in 1st trimester.

Q3c. Accessories for RIRS (SAQ)

- Hydrophilic wire passes; PTFE wire stays.
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